Block practice · More cells, DNA and cell division

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Locate DNA and chromosomes in a human cell

Sequence and describe the stages of mitosis

Describe genetic modification and gene therapy for human health

Explain and evaluate uses of embryonic and adult stem cells

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Practice workbook · Locate DNA and chromosomes in a human cell

Short items asking where DNA or chromosomes are found, or which molecule or structure carries genetic material.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-LOCATE-GENETIC-MATERIAL

1. Question 4 · part a.i · 1 mark

November 2023 Paper 1

Worked-example retry · Attempt part a.i only.

4 The mass of DNA in a body cell changes during the cell cycle.

(a) (i) Which structure in a cell contains DNA?

(1)

- ☒ **A** chromosome
- ☒ **B** endoplasmic reticulum
- ☒ **C** cell membrane
- ☒ **D** ribosome

(ii) Name **one** other type of nucleic acid.

(1)

.....

(iii) Give **three** differences between DNA and the nucleic acid that you have named in (a)(ii).

(3)

Differences	DNA	Named nucleic acid
1		
2		
3		

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1. Question 4 · part a.i · 1 mark · Working and answer

Working and answer

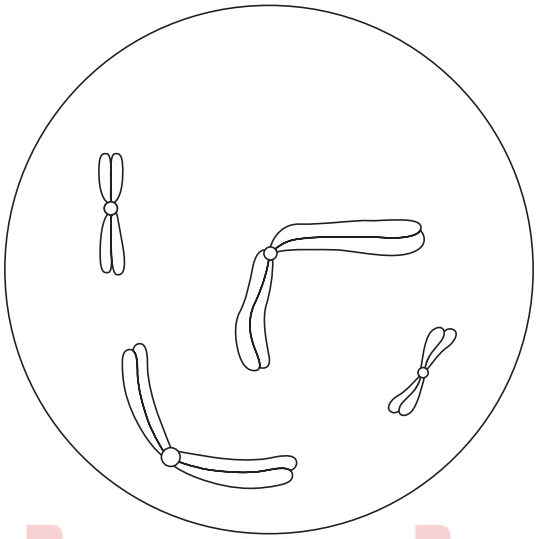
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2. Question 5 · part a.i · 1 mark

June 2021 Paper 2

Further practice · Attempt part a.i only.

5 The diagram shows a cell in the early stages of mitosis.



(a) (i) Which molecule is found in the chromosomes?

(1)

- ☒ **A** DNA
- ☒ **B** glucose
- ☒ **C** glycogen
- ☒ **D** mRNA

(ii) Draw a circle around two homologous chromosomes on the diagram.

(1)

(iii) State how the diagram shows that this is not a human cell.

(1)

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2. Question 5 · part a.i · 1 mark · Working and answer

Working and answer

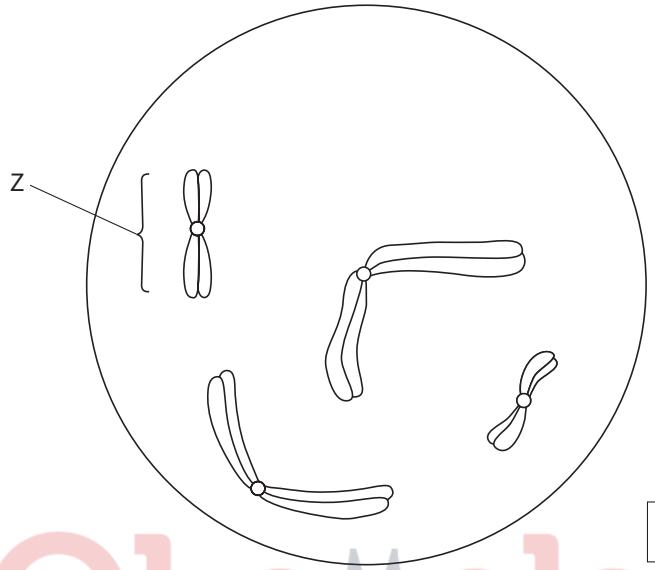
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3. Question 7 · part b.i · 1 mark

November 2021 Paper 1

Further practice · Attempt part b.i only.

7 The diagram shows a cell in early stages of mitosis.



(a) The four stages of mitosis are:

- anaphase
- metaphase
- prophase
- telophase

(i) Give these stages in the correct order.

(2)

.....

.....

(ii) Draw a diagram of the cell as it would appear at metaphase.

(3)

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P 6 7 0 6 0 R R A 0 1 6 2 0

3. Question 7 · part b.i · 1 mark

November 2021 Paper 1

Further practice · Attempt part b.i only.

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(b) (i) Which one of these molecules would be found in structure Z?

(1)

- ☒ **A** DNA
- ☒ **B** mRNA
- ☒ **C** tRNA
- ☒ **D** ribose sugar

(ii) Calculate the length of structure Z in micrometres.

[1 micrometre = 1×10^{-6} m]

(3)

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length of structure Z = ... micrometres

(c) Describe what is produced when a cell undergoes mitosis.

(2)

.....

.....

.....

.....

(Total for Question 7 = 11 marks)

3. Question 7 · part b.i · 1 mark · Working and answer

Working and answer

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4. Question 1 · part a.iv · 1 mark

June 2022 Paper 1

Further practice · Attempt part a.iv only.

1 (a) The diagram shows part of a molecule of DNA.



(i) Structure X is the backbone of the DNA molecule.

Which of these form part of the backbone of DNA?

(1)

- ☒ **A** amino acids
- ☒ **B** bases
- ☒ **C** proteins
- ☒ **D** sugars

(ii) What name is given to the structures labelled Y?

(1)

- ☒ **A** amino acids
- ☒ **B** bases
- ☒ **C** proteins
- ☒ **D** sugars

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4. Question 1 · part a.iv · 1 mark

June 2022 Paper 1

Further practice · Attempt part a.iv only.

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(iii) Describe how the two backbones of the DNA molecule are joined together.

(2)

(iv) State the name of the part of the cell where most of the DNA is found.

(1)

(b) DNA is a nucleic acid.

RNA is another type of nucleic acid.

(i) Give two differences between the structures of DNA and RNA.

(2)

1

2

(ii) There are different types of RNA.

Which of these are two types of RNA?

(1)

- ☒ **A** mRNA and pRNA
- ☒ **B** dRNA and tRNA
- ☒ **C** dRNA and pRNA
- ☒ **D** mRNA and tRNA

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4. Question 1 · part a.iv · 1 mark · Working and answer

Working and answer

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5. Question 3 · part a.i · 1 mark

June 2024 Paper 1

Further practice · Attempt part a.i only.

3 (a) Polydactyly is an inherited condition.

There are several faulty genes on different chromosomes that can cause polydactyly.

(i) State which part of a cell contains the most chromosomes.

(1)

(ii) Genes exist in alternative forms.

State the name given to an alternative form of a gene.

(1)

(iii) Give one symptom of polydactyly.

(1)

(b) Polydactyly is caused by the dominant form of the gene.

The table shows the genotypes of two parents, R and S.

Parent	Genotype
female (R)	Pp
male (S)	pp

(i) Name the term used to describe the genotype of parent R.

(1)

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P 7 4 6 2 6 R A 0 8 2 4

5. Question 3 · part a.i · 1 mark · Working and answer

Working and answer

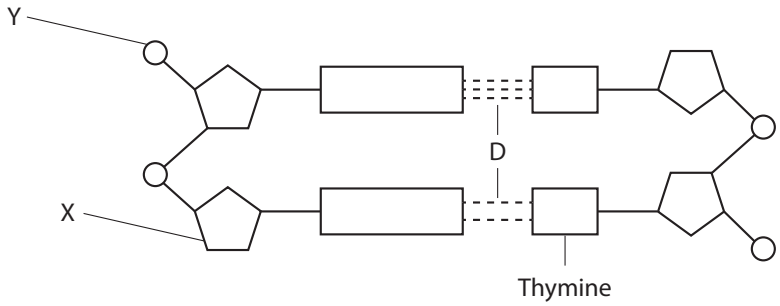
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6. Question 2 · part a.iv · 1 mark

June 2025 Paper 1

Further practice · Attempt part a.iv only.

2 The diagram shows part of a molecule of nucleic acid.



(a) (i) Name the parts labelled X and Y.

(2)

X .....

Y .....

(ii) Name the type of nucleic acid shown in the diagram.

(1)

.....

(iii) Use information from the diagram to justify your answer to part (ii).

(3)

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(iv) State where this nucleic acid is found in a human cell.

(1)

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P 7 8 7 6 8 A 0 4 2 0

6. Question 2 · part a.iv · 1 mark

June 2025 Paper 1

Further practice · Attempt part a.iv only.

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(b) (i) State what is labelled D in the diagram.

(1)

(ii) Describe the importance of D in maintaining the structure of the nucleic acid.

(3)

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(Total for Question 2 = 11 marks)

Working and answer

Practice workbook · Sequence and describe the stages of mitosis

Ordering prophase, metaphase, anaphase and telophase, describing chromosome behaviour in each, and drawing or recognising metaphase.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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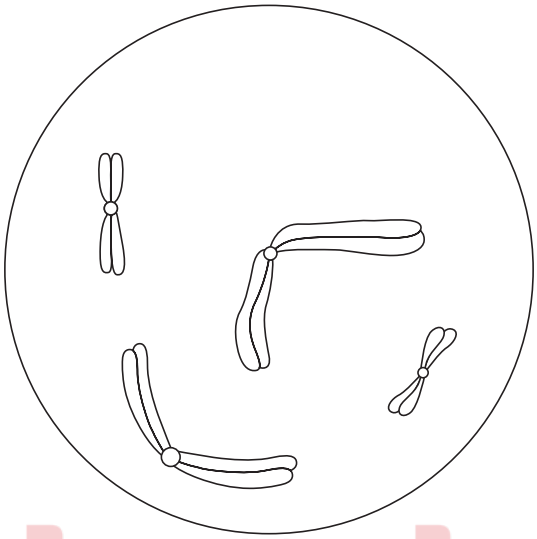
Type ID: HBI-SEQUENCE-MITOSIS-STAGES

1. Question 5 · part b · 5 marks

June 2021 Paper 2

Worked-example retry · Attempt part b only.

5 The diagram shows a cell in the early stages of mitosis.



(a) (i) Which molecule is found in the chromosomes? **(1)**

- ☒ **A** DNA
- ☒ **B** glucose
- ☒ **C** glycogen
- ☒ **D** mRNA

(ii) Draw a circle around two homologous chromosomes on the diagram. **(1)**

(iii) State how the diagram shows that this is not a human cell. **(1)**

.....

.....

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P 6 6 4 3 3 A 0 1 2 2 4

1. Question 5 · part b · 5 marks

June 2021 Paper 2

Worked-example retry · Attempt part b only.

(b) Describe what happens to the chromosomes during the four stages of mitosis.

(5)

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(Total for Question 5 = 8 marks)

1. Question 5 · part b · 5 marks · Working and answer

Working and answer

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2. Question 6 · part a.iv · 3 marks

November 2020 Paper 1R

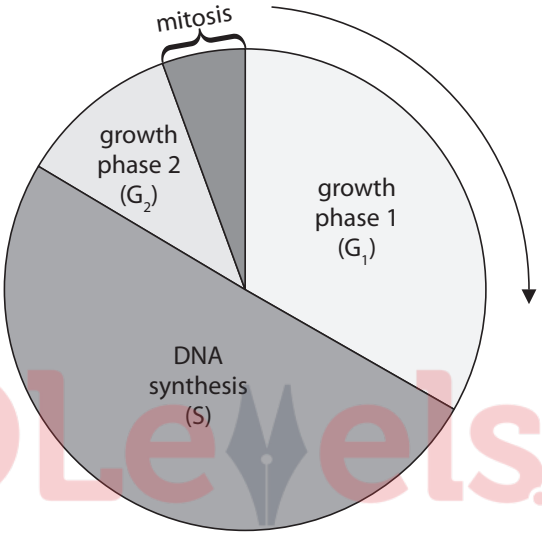
Further practice · Attempt part a.iv only.

6 (a) Cells in the body go through several cell cycles in their lifetime.

Each cell cycle is divided into different stages.

Different events take place in the cell during each stage.

The diagram shows one cell cycle of a body cell.



(i) One cell cycle takes 24 hours to complete.

The total time spent in G_1 and G_2 is ten hours.

Calculate the time spent in mitosis.

(2)

time = ... hours

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2. Question 6 · part a.iv · 3 marks

November 2020 Paper 1R

Further practice · Attempt part a.iv only.

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(ii) Each of the phases of the cell cycle prepare the cell for mitosis.

Suggest the events that take place in the growth phases of the cell cycle.

(2)

(iii) The cell cycle completes by producing two new body cells.

Describe the genetic features of the two cells produced by mitosis.

(2)

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(iv) The table lists some of the stages of mitosis.

Complete the table by giving details of the event that takes place in each stage.

(3)

Stage	Events
metaphase	
anaphase	
telophase	

2. Question 6 · part a.iv · 3 marks

November 2020 Paper 1R

Further practice · Attempt part a.iv only.

(b) Describe the role of DNA polymerase in the synthesis of new DNA molecules.

(2)

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(Total for Question 6 = 11 marks)

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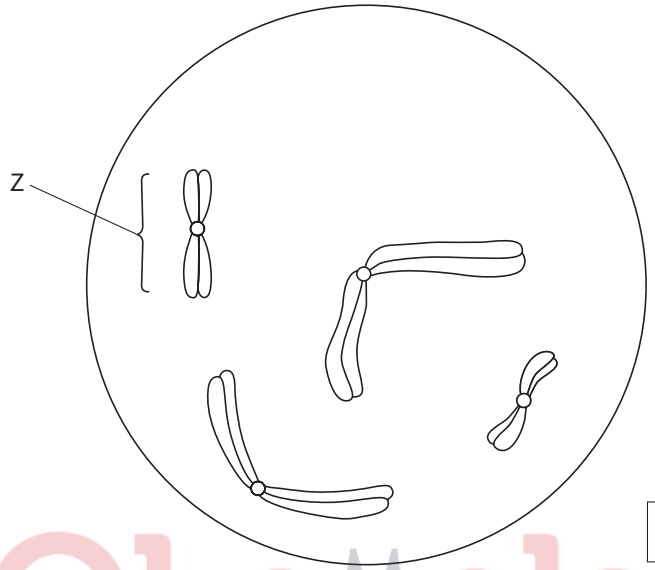
Working and answer

3. Question 7 · part a.i · 2 marks

November 2021 Paper 1

Further practice · Attempt part a.i only.

7 The diagram shows a cell in early stages of mitosis.



(a) The four stages of mitosis are:

- anaphase
- metaphase
- prophase
- telophase

(i) Give these stages in the correct order.

(2)

.....

.....

(ii) Draw a diagram of the cell as it would appear at metaphase.

(3)

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3. Question 7 · part a.i · 2 marks · Working and answer

Working and answer

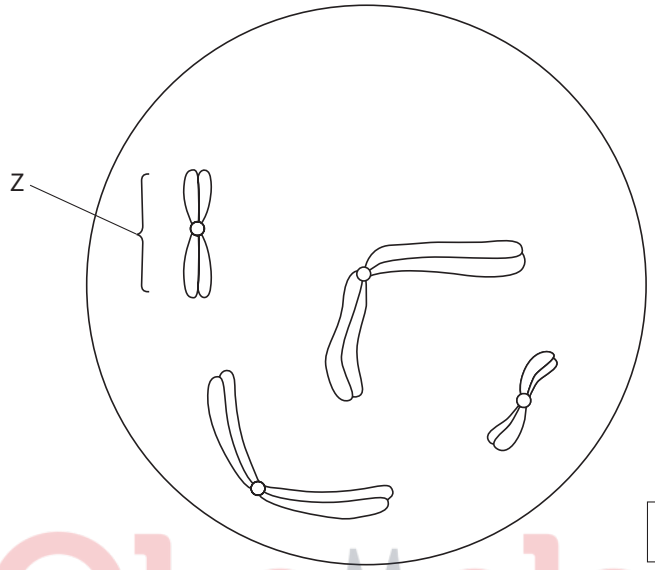
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4. Question 7 · part a.ii · 3 marks

November 2021 Paper 1

Further practice · Attempt part a.ii only.

7 The diagram shows a cell in early stages of mitosis.



(a) The four stages of mitosis are:

- anaphase
- metaphase
- prophase
- telophase

(i) Give these stages in the correct order.

(2)

.....

.....

(ii) Draw a diagram of the cell as it would appear at metaphase.

(3)

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P 6 7 0 6 0 R R A 0 1 6 2 0

4. Question 7 · part a.ii · 3 marks

November 2021 Paper 1

Further practice · Attempt part a.ii only.

(b) (i) Which one of these molecules would be found in structure Z?

(1)

- ☒ **A** DNA
- ☒ **B** mRNA
- ☒ **C** tRNA
- ☒ **D** ribose sugar

(ii) Calculate the length of structure Z in micrometres.

[1 micrometre = 1×10^{-6} m]

(3)

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length of structure Z = ... micrometres

(c) Describe what is produced when a cell undergoes mitosis.

(2)

.....

.....

.....

.....

(Total for Question 7 = 11 marks)

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4. Question 7 · part a.ii · 3 marks · Working and answer

Working and answer

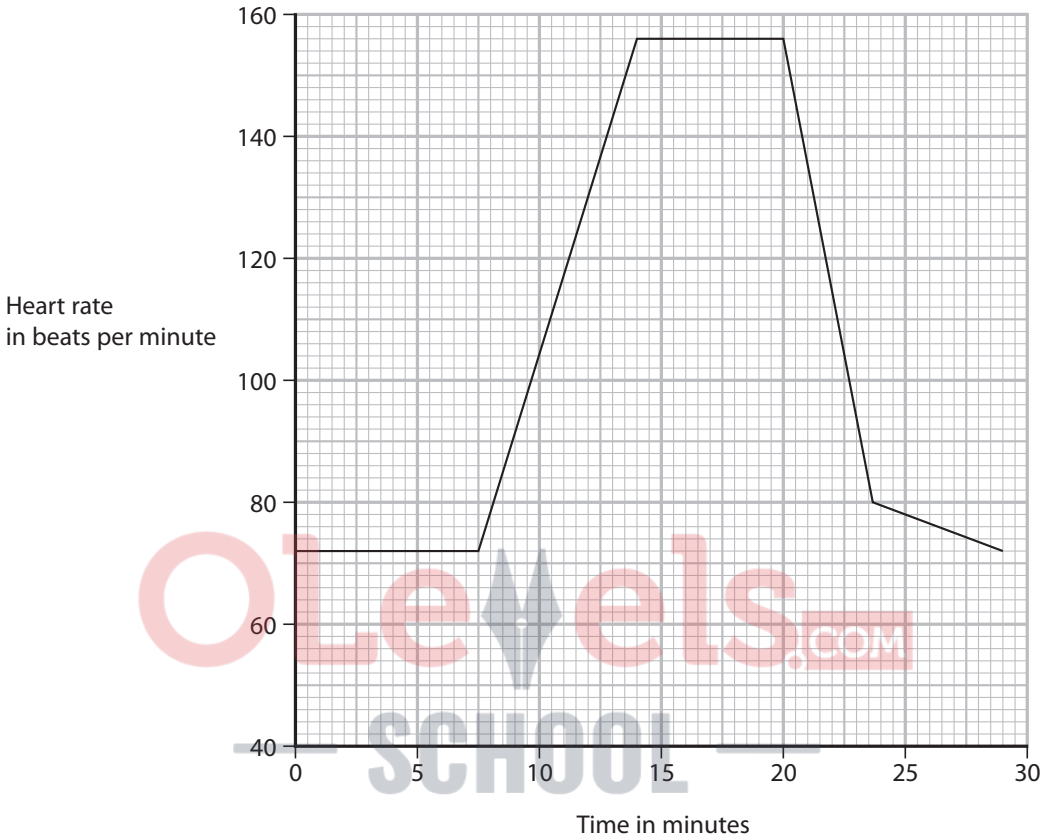
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5. Question 2 · part b.ii · 1 mark

June 2022 Paper 1

Further practice · Attempt part b.ii only.

2 (a) The graph shows how a person's heart rate changes with exercise.



(i) Determine the difference between the maximum heart rate and the resting heart rate.

(2)

difference = ... beats per minute

(ii) Which of these is the time taken for this person to recover from the exercise?

(1)

- ☒ **A** 4 minutes
- ☒ **B** 6 minutes
- ☒ **C** 9 minutes
- ☒ **D** 15 minutes

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5. Question 2 · part b.ii · 1 mark

June 2022 Paper 1

Further practice · Attempt part b.ii only.

(iii) Complete the passage about what happens in the body during exercise using the correct words.

(4)

During exercise ..... gas moves from the lungs into the blood by a process called ..... . This gas is needed by cells for ..... respiration which releases energy. Also during exercise, the amount of urine produced ..... as water is lost in sweat.

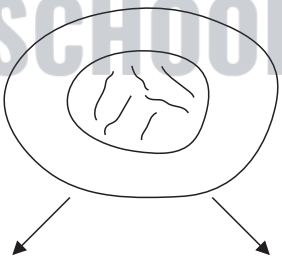
(b) (i) During exercise, body cells get damaged.

Body cells are replaced by a process called mitosis.

The diagram represents one body cell but shows only 6 of the 46 chromosomes.

Complete the diagram to show the new body cells produced from this cell by mitosis.

(2)



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P 7 0 9 7 3 A 0 6 2 4

5. Question 2 · part b.ii · 1 mark

June 2022 Paper 1

Further practice · Attempt part b.ii only.

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(ii) Mitosis takes place in four stages.

Which of these gives the stage of mitosis where chromosomes align at the equator of the cell?

(1)

- ☒ **A** anaphase
- ☒ **B** metaphase
- ☒ **C** prophase
- ☒ **D** telophase

(Total for Question 2 = 10 marks)

Working and answer

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6. Question 2 · part b.i · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.i only.

2 (a) Cells are made up of different components. Each component has a particular function.

Identify the cell component that transports substances inside the cell.

(1)

- ☒ **A** cell membrane
- ☒ **B** cytoplasm
- ☒ **C** endoplasmic reticulum
- ☒ **D** ribosomes

(b) Mitosis is a type of cell division.

A student uses a light microscope to view a cell undergoing four stages in mitosis.

The diagrams, W, X, Y and Z, show the student's observations.



W X Y Z

(Source: https://www.pinterest.co.uk/pin/812547957739062607/)

(i) Give the letters to show the correct order of the diagrams in the process of mitosis.

(1)

.....

(ii) State one function of mitosis in body cells.

(1)

.....

.....

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6. Question 2 · part b.i · 1 mark · Working and answer

Working and answer

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Practice workbook · Describe genetic modification and gene therapy for human health

Engineering bacteria or yeast to make human insulin, Golden Rice for vitamin A, and gene therapy that delivers a working allele into body cells with a viral vector.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-DESCRIBE-GENETIC-MODIFICATION

1. Question 6 · part whole · 6 marks

June 2022 Paper 1

Worked-example retry · Attempt part whole only.

6 Describe how bacteria can be genetically modified to produce human insulin.

(6)

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(Total for Question 6 = 6 marks)

1. Question 6 · part whole · 6 marks · Working and answer

Working and answer

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2. Question 6 · part c · 3 marks

January 2023 Paper 2

Further practice · Attempt part c only.

6 (a) Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

In 2012, Shinya Yamanaka, a Japanese research scientist, won a Nobel Prize for his work on stem cells. He demonstrated that it was possible to take a sample of blood or skin cells from a person and induce these cells to have the properties of stem cells.

In the past, stem cell research scientists needed to obtain human embryonic stem cells from embryos produced by IVF. This required the agreement of those people involved. Embryonic cells can continually divide and can become any human body cell. Scientists have been exploring the use of embryonic stem cells as medical treatments for many different diseases and conditions.

Using the work of Yamanaka and his team, scientists can now produce cells called induced pluripotent stem cells (iPS cells). Cells from the patient are treated with four factors which control the expression of genes. This produces iPS cells from the patient's own cells, which can then be used in stem cell therapies.

This technique is easy to replicate but there are concerns that unlicensed clinics are offering stem cell treatments. Many of these treatments are expensive and success rates are not proven. In addition, stem cells have the potential to continue to divide in the body. Many of the diseases and conditions that stem cell therapies could be used to treat have no other effective treatments. This makes stem cell therapy a desirable option.

(i) Describe the benefits of producing iPS cells from a patient's own cells.

(3)

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2. Question 6 · part c · 3 marks

January 2023 Paper 2

Further practice · Attempt part c only.

(ii) Give two possible risks of using stem cells to treat diseases and medical conditions.

(2)

1 .....

.....

.....

2 .....

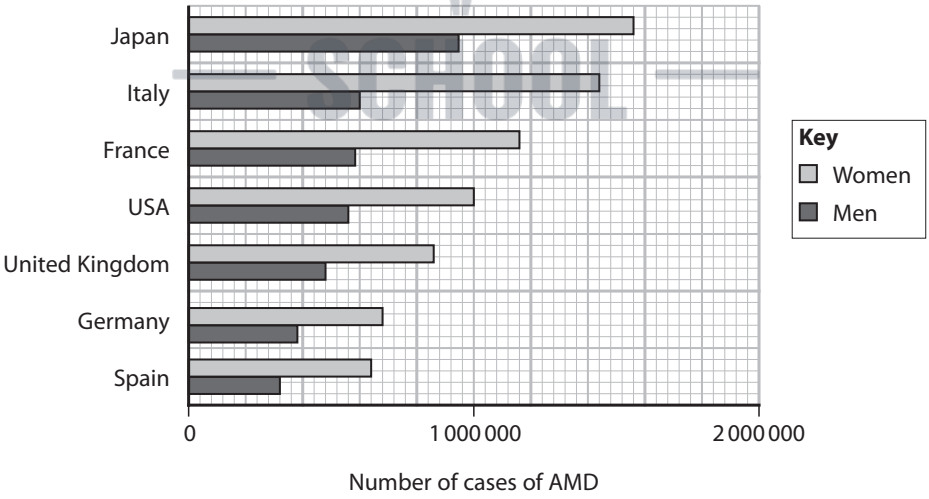
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(b) One condition that stem cells could be used to treat is age-related macular degeneration (AMD).

This is the main cause of vision loss in people over the age of 50.

The graph shows the number of people affected by AMD in some countries.



(i) Calculate the difference between the number of women and the number of men affected by AMD in the United Kingdom.

(2)

difference = ...

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P 7 2 6 0 2 A 0 1 9 2 8

Turn over ▶

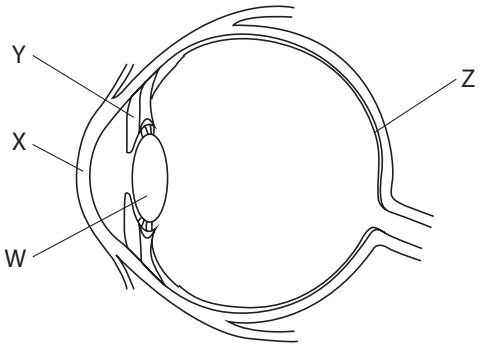
2. Question 6 · part c · 3 marks

January 2023 Paper 2

Further practice · Attempt part c only.

(ii) The diagram shows a human eye.

AMD affects the retina of the eye.



Which of these structures is affected by AMD?

(1)

- ☒ **A** structure W
- ☒ **B** structure X
- ☒ **C** structure Y
- ☒ **D** structure Z

(c) A deficiency of vitamin A can also cause blindness.

A genetically engineered rice product can be used to prevent deficiency of vitamin A.

Describe how genetic engineering is used to produce this rice product.

(3)

(Total for Question 6 = 11 marks)

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2. Question 6 · part c · 3 marks · Working and answer

Working and answer

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3. Question 7 · part e · 2 marks

June 2025 Paper 2R

Further practice · Attempt part e only.

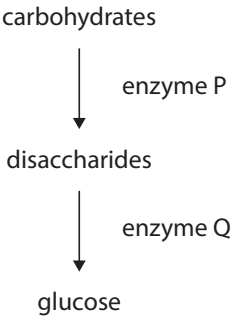
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7 (a) The diagram shows how carbohydrates are broken down to glucose by two different enzymes, enzyme P and enzyme Q.

These enzymes are found in different regions of the digestive system.



Name enzyme P.

(1)

.....

(b) People with diabetes can use glucose testing strips to test their urine.

Describe how enzymes are used in glucose testing strips.

(2)

.....

.....

.....

.....

(c) People with type 1 diabetes are unable to produce hormones to control their blood glucose levels.

Name the organ responsible for producing hormones to control levels of glucose in the blood.

(1)

.....

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3. Question 7 · part e · 2 marks

June 2025 Paper 2R

Further practice · Attempt part e only.

(d) Insulin is a hormone used to treat people with diabetes.

Until recently insulin was obtained from animals for human use.

Suggest why this type of insulin can cause health complications in people with diabetes.

(2)

(e) Describe how yeast cells can be modified to produce the human form of insulin.

(2)

(Total for Question 7 = 8 marks)

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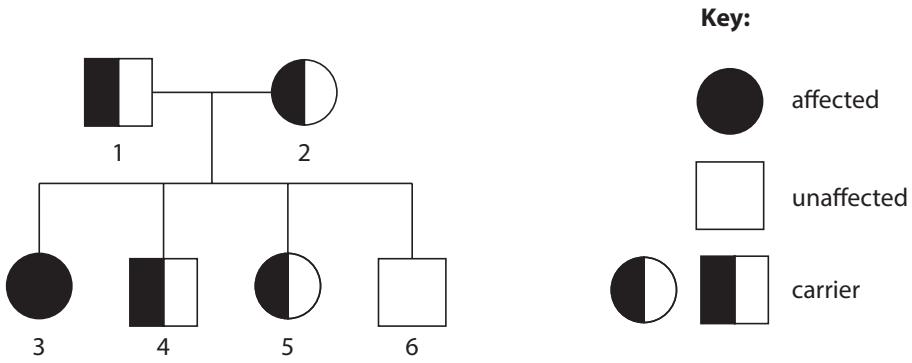
Working and answer

4. Question 8 · part b · 3 marks

June 2024 Paper 2

Further practice · Attempt part b only.

8 (a) The diagram shows the inheritance of cystic fibrosis in a family.



(Source: Adapted from https://cysticfibrosiszaw2.weebly.com/)

(i) Which individual is homozygous recessive?

(1)

- ☒ **A** 1
- ☒ **B** 3
- ☒ **C** 5
- ☒ **D** 6

(ii) How many individuals have a heterozygous genotype?

(1)

- ☒ **A** 1
- ☒ **B** 2
- ☒ **C** 3
- ☒ **D** 4

(iii) Explain why carriers of cystic fibrosis do not have symptoms of the disease.

(2)

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P 7 4 6 3 0 A 0 2 2 2 4

4. Question 8 · part b · 3 marks

June 2024 Paper 2

Further practice · Attempt part b only.

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(iv) Draw a genetic diagram to show how two parents can have a 1 in 4 chance of having a child with cystic fibrosis.

Use F for the non-cystic fibrosis allele and f for the cystic fibrosis allele.

(4)

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(b) Describe how gene therapy can be used to treat cystic fibrosis.

(3)

(Total for Question 8 = 11 marks)

4. Question 8 · part b · 3 marks · Working and answer

Working and answer

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Practice workbook · Explain and evaluate uses of embryonic and adult stem cells

Why unspecialised stem cells are useful, treating disease by differentiation, and weighing benefits, risks and ethics of embryonic, adult and induced cells.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-EVALUATE-STEM-CELL-USE

1. Question 8 · part whole · 6 marks

June 2022 Paper 1

Worked-example retry · Attempt part whole only.

8 Discuss the advantages and disadvantages of the use of embryonic and adult stem cells in medical research.

(6)

embryonic stem cells

adult stem cells

(Total for Question 8 = 6 marks)

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1. Question 8 · part whole · 6 marks · Working and answer

Working and answer

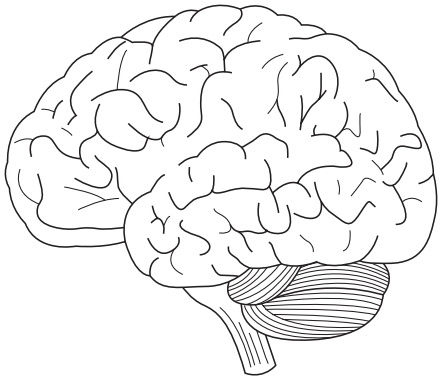
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2. Question 7 · part c · 3 marks

June 2019 Paper 1

Further practice · Attempt part c only.

7 (a) The diagram shows a human brain.



Add lines labelled X, Y and Z to the diagram to show the areas of the brain that control these functions.

(3)

X voluntary actions

Y balance

Z breathing rate

(b) Parkinson's disease affects the cells in the brain that help to control body movement.

The affected cells are unable to communicate effectively with neurones that cause muscles to contract.

(i) Name the type of neurone that causes muscles to contract.

(1)

(ii) Give one difference between neurones that cause muscles to contract and neurones that transmit nerve impulses from receptor organs.

(1)

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2. Question 7 · part c · 3 marks

June 2019 Paper 1

Further practice · Attempt part c only.

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(iii) Explain how one neurone communicates with another neurone.

(3)

(c) Recent scientific research into the treatment of Parkinson's disease has involved the use of stem cells.

Explain how stem cells could be used to reduce the symptoms of Parkinson's disease.

(3)

(Total for Question 7 = 11 marks)

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2. Question 7 · part c · 3 marks · Working and answer

Working and answer

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3. Question 6 · part c · 1 mark

June 2023 Paper 2R

Further practice · Attempt part c only.

6 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Pregnancy begins with the formation of a zygote. This stage is known as conception. The zygote develops into an embryo. By day five the embryo is transformed into a blastocyst, a hollow ball of about 100 genetically identical cells. These cells have not yet differentiated and they can provide a source of embryonic stem cells for use in medical research. Embryos produced following IVF treatment can be used as a source of embryonic stem cells.

In the second week following conception, the blastocyst begins the process of implantation and releases a hormone, called hCG. This hormone maintains the corpus luteum, a structure formed in the ovary following ovulation. After implantation the amniotic sac forms, which is where the blastocyst and later the fetus develop.

In the fourth and fifth week after conception, the umbilical cord forms. This connects the embryo to the placenta by week eight. The placenta is not fully developed until week ten when it begins to produce progesterone and oestrogen. All organ systems are present at this time in pregnancy.

Fetal development begins at about month four and it is possible to distinguish males from females at this time. In males the testes, which develop in the abdominal cavity, release testosterone. Female characteristics are produced in the fetus if testosterone is not released at the stage.

Hormones are important in starting labour and the birth process, about nine months after conception. The cervix dilates, with the head of the baby pushing against the cervix to help the process of dilation. Once the child is born and breathing normally, the umbilical cord is tied and cut, separating the child from the placenta.

(a) Describe the process that occurs at conception. (lines 1 and 2)

(2)

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3. Question 6 · part c · 1 mark

June 2023 Paper 2R

Further practice · Attempt part c only.

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(b) Describe the process that forms a blastocyst. (lines 2 to 4)

(4)

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(c) Which is a reason why cells taken from a blastocyst can be useful in scientific research?

(1)

- ☒ **A** the cells are already differentiated
- ☒ **B** the cells are able to differentiate into both sex cells and body cells
- ☒ **C** the cells are able to differentiate into any type of sex cell
- ☒ **D** the cells are able to differentiate into any type of body cell

3. Question 6 · part c · 1 mark

June 2023 Paper 2R

Further practice · Attempt part c only.

(d) Explain why it is important that hCG maintains the corpus luteum. (lines 7 to 9)

(2)

(e) Explain the importance of the development that takes place in the fourth and fifth week after conception. (lines 12 and 13)

(2)

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(f) Suggest the role of testosterone in the development of the fetus. (lines 17 to 19)

(1)

(g) Which change occurs during the birth process? (line 20)

(1)

- ☒ **A** oestrogen levels decrease
- ☒ **B** progesterone levels increase
- ☒ **C** release of oxytocin from the pituitary gland
- ☒ **D** release of hCG from the placenta

(Total for Question 6 = 13 marks)

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3. Question 6 · part c · 1 mark · Working and answer

Working and answer

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4. Question 6 · part a.i · 3 marks

January 2023 Paper 2

Further practice · Attempt part a.i only.

6 (a) Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

In 2012, Shinya Yamanaka, a Japanese research scientist, won a Nobel Prize for his work on stem cells. He demonstrated that it was possible to take a sample of blood or skin cells from a person and induce these cells to have the properties of stem cells.

In the past, stem cell research scientists needed to obtain human embryonic stem cells from embryos produced by IVF. This required the agreement of those people involved. Embryonic cells can continually divide and can become any human body cell. Scientists have been exploring the use of embryonic stem cells as medical treatments for many different diseases and conditions.

Using the work of Yamanaka and his team, scientists can now produce cells called induced pluripotent stem cells (iPS cells). Cells from the patient are treated with four factors which control the expression of genes. This produces iPS cells from the patient's own cells, which can then be used in stem cell therapies.

This technique is easy to replicate but there are concerns that unlicensed clinics are offering stem cell treatments. Many of these treatments are expensive and success rates are not proven. In addition, stem cells have the potential to continue to divide in the body. Many of the diseases and conditions that stem cell therapies could be used to treat have no other effective treatments. This makes stem cell therapy a desirable option.

(i) Describe the benefits of producing iPS cells from a patient's own cells.

(3)

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4. Question 6 · part a.i · 3 marks

January 2023 Paper 2

Further practice · Attempt part a.i only.

(ii) Give two possible risks of using stem cells to treat diseases and medical conditions.

(2)

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2 .....

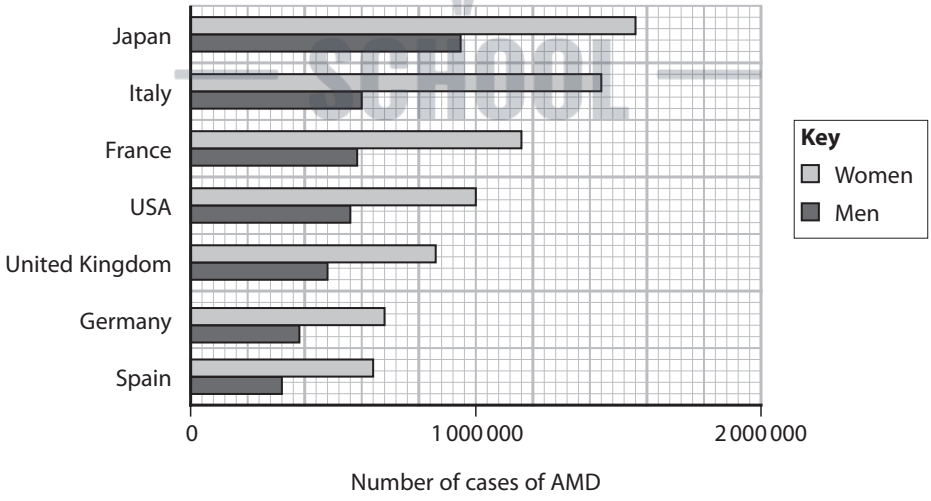
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(b) One condition that stem cells could be used to treat is age-related macular degeneration (AMD).

This is the main cause of vision loss in people over the age of 50.

The graph shows the number of people affected by AMD in some countries.



(i) Calculate the difference between the number of women and the number of men affected by AMD in the United Kingdom.

(2)

difference = ...

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4. Question 6 · part a.i · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 6 · part a.ii · 2 marks

January 2023 Paper 2

Further practice · Attempt part a.ii only.

6 (a) Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

In 2012, Shinya Yamanaka, a Japanese research scientist, won a Nobel Prize for his work on stem cells. He demonstrated that it was possible to take a sample of blood or skin cells from a person and induce these cells to have the properties of stem cells.

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(i) Describe the benefits of producing iPS cells from a patient's own cells.

(3)

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5. Question 6 · part a.ii · 2 marks

January 2023 Paper 2

Further practice · Attempt part a.ii only.

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(ii) Give two possible risks of using stem cells to treat diseases and medical conditions.

(2)

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2 .....

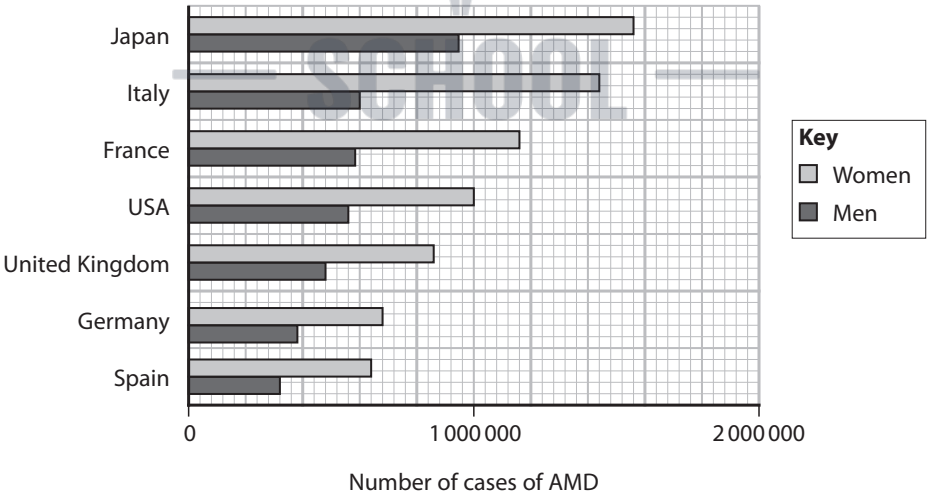
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(2)

difference = ...

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Turn over ▶

5. Question 6 · part a.ii · 2 marks · Working and answer

Working and answer

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